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$$\begin{aligned} & \frac{(1+i)^2}{(1-i)^3} + \frac{(1-i)^2}{(1+i)^3} \\ &= \frac{1+2i+i^2}{(1-i)(1-2i+i^2)} + \frac{1-2i+i^2}{(1+i)(1+2i+i^2)} \\ &= \frac{2i}{(1-i)(-2i)} + \frac{-2i}{(1+i)(2i)} \\ &= \frac{2i}{-2-2i} + \frac{-2i}{-2+2i} \\ &= \frac{i}{-1-i} + \frac{-i}{-1+i} \\ &= \frac{i \cdot (-1+i)}{(-1-i) \cdot (-1+i)} + \frac{-i \cdot (-1-i)}{(-1+i) \cdot (-1-i)} \\ &= \frac{-i+i^2}{1+i-i-i^2} + \frac{i+i^2}{1+i-i-i^2} \\ &= \frac{2i^2}{-i^2+1} \\ &= \frac{-2}{2} \\ &= -1 \end{aligned}$$

References